SWE645 – Assignment 2

Overview

This homework assignment may be completed individually or in a group of up to four (4) students. If you are working in a group, you must clearly identify each member's contributions in the documentation.

Assignment Requirements

You are required to complete the following tasks:

1. **Containerization**
   - Containerize the web application you developed in Homework 1 – Part 2 using Docker
   - Tag the Docker image and push the tagged image to Docker Hub (https://hub.docker.com).

2. **Kubernetes Deployment**
   - Create Deployment and Service manifest files
   - Use the `kubectl` command-line tool to deploy the containerized application to Kubernetes.
   - Your baseline configuration must include at least three (3) pods running at all times for resiliency.
     - If one of the pods is deleted, another pod should be created automatically in its place.
   - You may use one of the following options:
     - Rancher to set up a Kubernetes cluster (https://rancher.com/docs)
     - A managed Kubernetes service, such as:
       - Amazon EKS (AWS)
       - Google GKE (GCP)

3. **CI/CD Pipeline**
   - Establish a CI/CD pipeline that includes:
     - A GitHub repository for source code management
     - Jenkins for:
       - Automated builds
       - Automated deployment to Kubernetes
   - Re-deploy your containerized application on Kubernetes cluster using the CI/CD pipeline.

Submission Instructions

All submissions must be made through Canvas. Please submit a single zipped package containing:

- Source code
- Configuration files, including:
    - Dockerfile
    - Jenkinsfile
    - Kubernetes YAML files
- Any additional scripts or supporting files used

Documentation

Include a detailed documentation file (PDF or Word) that explains:

- What you built
- How it works
- Installation and setup instructions
- CI/CD pipeline details
- Kubernetes deployment details

Video Demonstration

You must also include a video recording with voice-over that demonstrates:

- The setup process
- Deployment on K8s cluster using kubectl
- The CI/CD pipeline execution. For a working demonstration of the CI/CD pipeline, you are supposed to show whether changing the git repo triggers the CI/CD pipeline and if you can see the changes on the updated website.
- The running application on Kubernetes
- The AWS (or cloud) URL of the deployed Kubernetes application used during the recording

The documentation and video should be detailed enough for the TA or instructor to replicate your work. This material may also serve as a reference for future assignments.

Important Notes

- Late submissions incur a 10% penalty per week.
- Assignments will not be accepted after being more than two (2) weeks late.
- Ensure your name (or group member names) appear on all programming artifacts.
- Each source file must include 1–2 sentences comments at the top describing its purpose.

Grading Criteria

Criteria	Points
Meets functional requirements	60
Runs without errors	15
Submission package, documentation, and video quality	25
Total	**100**

Instant Point Deductions

Points may be deducted immediately if:

- Source code or configuration files are missing
- Detailed documentation is not included
- Video recording is missing or does not include voice-over
- Instructions are incomplete or unclear
- The assignment cannot be executed due to missing steps or information